

Mixed Opinions, Positive Returns: Explaining Nike Stock Returns
Using LLM-Based Sentiment Analysis of Athlete Media Coverage

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Abstract: This paper examines whether daily media sentiment about Nike-endorsed athletes helps explain Nike stock returns, bridging a gap in the literature that has relied on discrete event studies. Using daily stock returns and 8,000+ New York Times articles covering 16 athletes from 1984–2025 we use looped LLM sentiment ratings on a trinomial to construct daily continuous sentiment variables. Sentiment variables are constructed using an exponential decay weighting scheme, with decay window and persistence rate tuned via 20-fold time series cross-validation. Our main finding is that sentiment volatility (disagreement across articles on a given day) is a significant predictor of Nike returns and is robust across LLM temperatures, market controls, and year fixed effects. We also test this by seeing if Nike player sentiment variables have significant effects on non-Nike stocks. To this we find no significant effects and unstable tuning results, which suggests we are truly capturing something about Nike and not just some general market. We also find when sentiment variables are split by player, markets react differently to different players, and a career-period analysis of Kobe Bryant shows this heterogeneity stems from player specific market sensitivity rather than differing reactions to negative news. Finally we show that reliable results cannot be produced with one-shot prompting and rather our looped LLM structure is needed to produce structured and reliable results.

Keywords: Celebrity endorsements, LLM sentiment analysis, stock returns, Nike, athlete media coverage, sentiment dispersion

JEL Codes: G14, Z23, M37

1 Introduction

1.1 Hook and Research Question

In 1984, Nike signed rookie basketball player Michael Jordan to a five-year endorsement with a base pay of \$2.5 million (Kunkel, 2023). This deal revolutionized the industry even further as it paved the way for royalty deals and let athletes and the brands become one in the consumers mind as the Air Jordan sales generated \$126 million in its first year alone. With Jordan's and other athletes' value to Brands so obvious one could only imagine the negative and positive implications that come from athlete scandals or successes. After all, part of the Air Jordan success came from that Micheal was one of the greatest young talents at the time.

Thus in this paper we look for answer the following questions. Does the public view and performance of a player help explain stock returns of that player's sponsored brands? What direction are these effects and are we able to clearly state the the effect is coming from the player or rather a more sector wide signal? Answers to this will help us in turn answer what types of players the market reacts more sensitively too.

1.2 Literature Review

Existing literature begins with Agrawal & Kamakura (1995) who perform a event study of 110 celebrity endorsement announcements. The find significant positive abnormal returns of 0.44% on announcement day which implies that markets view celebrity deals as worthy investments.

Elberse & Verleun (2012) extend this line of thinking by examining not just stock returns but also brand-level sales. They study 341 athlete endorsement deals across a wide range of consumer goods categories. They find that signing an endorser increases weekly brand sales by approximately 4% on average, with sales and stock returns rising further with each major athletic achievement by the endorser. However, they show that these sales effects exhibit decreasing returns over time, which flags a need for ongoing athlete/celebrity-specific information rather than just signing days.

In the remaining Literature we see multiple papers use the professional golfer Tiger Woods to study how a sponsored athlete effects a brand. This is due to how he is undeniably the greatest golfer of his generation and how he had severe scandals during his time with Nike.

Chung, Derdenger & Srinivasan (2013) further strengthen the findings of Elberse & Verleun (2012) at the product level as they show that 'Tiger Woods' tournament performance significantly affected Nike golf ball

sales even after controlling for product, price, and time fixed effects.

A event study performed by Knittel & Stango (2014) found that the Tiger Woods scandal caused sponsors to lose over 2% of market value in 10-15 trading days. However Hood (2012) rebuttals to the early working paper version of this paper by showing that the results vary based on estimation and event windows. While Hood’s paper was framed more as a criticism of event window studies Knittel & Stango (2014) justify their window sizes with Google search search intensity data.

Given that we must account player’s popularity/performance across their career the most natural next step is to include media sentiments about athletes. In this Literature we first saw Tetlock (2007) establish that media text can be used to predict stock returns. They do this with a pessimism measure from the Wall Street Journal’s “Abreast of the Market” column using principal components analysis of automated word counts from the Harvard IV-4 psychosocial dictionary.

More modern methodology have looked to off-the-shelf and fine-tuned Large Language Models to perform the sentiment analysis. Kirtac, K. & Germano, G. (2024) achieve a Sharpe ratio of 3.05 and 355% return over two years with a fine-tuned OPT model from Hugging Face, versus 0.91% for the standard dictionary-based strategy.

Most similar to our work, Lopez-Lira & Tang (2024) use ChatGPT with zero-shot prompting (no fine-tuning) to classify 67,586 news headlines as good, bad, or unknown for the stock price. Their trinomial sentiment scoring is similar to our +1/0/-1 approach. They use panel regression with firm and date fixed effects. GPT-4 delivers a long-short Sharpe ratio of 3.8, while smaller models (GPT-1, BERT) show no significant predictability.

1.3 Research Gap and Motivations

The gap within the literature is then quite clear. While we see evidence of athletes effecting stock returns there is no literature that goes beyond a set of discrete events. By incorporating modern LLMs for sentiment analysis of daily NYT articles we will bridge this gap.

Thus our research will fill this gap and contribute to related literature in three main ways. First, we provide the first daily/continuous link between athlete media sentiment and sponsor stock returns. Second, we will provide empirical evidence that the volatility of the sentiment matters more then the level of sentiment. This is to say that markets react more to a change in opinions about a player. Lastly we look to answer the question of types of players have qualities that make stock returns sensitive to them. Thus similar algorithms could then be constructed to exploit this phenomenon going forward.

2 Model

2.1 Sentiment Variables

Let r_t denote the daily return on trading day t . For article j published on date d_j , its weight on day t is:

$$w_{jt} = z^{t-d_j} \cdot \mathbf{1}\{0 < t - d_j \leq w\} \quad (1)$$

where $w \in \{5, 10, \dots, 100\}$ is the decay window (trading days) and $z \in (0.01, 0.02, \dots, 1.00]$ is the persistence rate. The five sentiment regressors are:

$$\text{sent}_t = \frac{\sum_j w_{jt} s_j}{\sum_j w_{jt}}, \quad s_j \in \{-1, 0, +1\} \quad (2)$$

$$\text{freq}_t = \sum_j w_{jt} \quad (3)$$

$$\text{sent_sd}_t = \sqrt{\frac{\sum_j w_{jt} (s_j - \text{sent}_t)^2}{\sum_j w_{jt}}} \quad (4)$$

$$\text{relev}_t = \frac{\sum_j w_{jt} (1 - n_j^{\text{NA}}/25)}{\sum_j w_{jt}} \quad (5)$$

$$\text{no_news}_t = \mathbf{1}\{\text{freq}_t = 0\} \quad (6)$$

Equations (2), (4), and (5) are defined only when $\text{freq}_t > 0$; on no-news days they are set to zero and equation (6) absorbs the level shift. Relevance is dropped in the single-run models since each article receives exactly one rating. By construction the sentiment variables do not include current t information. It is also vital to point out the equation (4) is not a measure of volatility among the 25 runs but rather a measure of the disagreement between articles on a given day.

2.2 Baseline Model

$$r_t = \alpha + \beta_1 r_{t-1} + \beta_2 \text{sent}_t + \beta_3 \text{freq}_t + \beta_4 \text{sent_sd}_t + \beta_5 \text{relev}_t + \beta_6 \text{no_news}_t + \varepsilon_t \quad (7)$$

2.3 Parameter selection

(w, z) are chosen via $K=20$ -fold time-series cross-validation. Under the expanding scheme, fold k trains on observations $1, \dots, b_k$ and tests on $b_k+1, \dots, b_{k+1}$; under the rolling scheme with memory m , training uses only the m most recent blocks. The fold- k out-of-sample RMSE is:

$$\text{RMSE}_k(w, z) = \sqrt{\frac{1}{n_k} \sum_{t \in T_k} (\hat{r}_t - r_t)^2} \quad (8)$$

where n_k is the number of test observations in fold k . The optimal pair minimizes a composite rank:

$$(w^*, z^*) = \arg \min_{(w, z)} [0.50 \cdot \text{rank}(\overline{\text{RMSE}}) + 0.25 \cdot \text{rank}(\text{AIC}) + 0.25 \cdot \text{rank}(\text{BIC})] \quad (9)$$

Half weight goes to out-of-sample fit; the remaining half is split equally between AIC and BIC. Note that in this case AIC and BIC have the same ranking.

2.4 Model Extensions

The market control model adds the contemporaneous S&P 500 return:

$$r_t = \alpha + \beta_1 r_{t-1} + \gamma \text{MktRet}_t + \sum_{k=2}^6 \beta_k X_{kt} + \varepsilon_t \quad (10)$$

The year fixed effects model absorbs annual variation:

$$r_t = \alpha + \beta_1 r_{t-1} + \sum_y \delta_y \mathbf{1}\{\text{year}_t = y\} + \sum_{k=2}^6 \beta_k X_{kt} + \varepsilon_t \quad (11)$$

where X_{kt} denotes the five sentiment regressors in both cases.

2.5 All-players model

Each player $p \in \mathcal{P}$ contributes their own sentiment variables, estimated jointly:

$$r_t = \alpha + \beta_1 r_{t-1} + \sum_{p \in \mathcal{P}} \sum_k \beta_{kp} X_{kpt} + \varepsilon_t \quad (12)$$

2.6 Kobe Bryant model

The career is partitioned into periods $\ell \in \{\text{pre-Nike, Nike splits, post-death}\}$ and sentiment effects vary by period:

$$r_t = \alpha + \beta_1 r_{t-1} + \sum_{\ell} (\beta_{1\ell} \text{sent}_t + \beta_{2\ell} \text{sent_sd}_t) \cdot \mathbf{1}\{\text{period}_t = \ell\} + \gamma^\top \mathbf{c}_t + \varepsilon_t \quad (13)$$

where $\mathbf{c}_t$ contains frequency, relevance, and no-news each interacted with period indicators, plus the uninteracted no-news

All standard errors are HC3.

3 Data

All of our data falls under two main categories; stock returns or news data. In terms stock returns we work with Nike returns mainly. We also include four other stocks that are of interest due to sector or athlete overlap. We include the S&P 500 as well for Market controls. While we will merge these two sources to create our data we can first show them individually.

3.1 Stock Returns

The summary of our stock return data and its stationary is seen below. All stock data is pulled from yahoo finance.

Table 1: Summary Statistics: Daily Stock Returns

Stock	N	Mean	Std. Dev.	Skewness	Ex. Kurtosis	ADF p-value
Nike	8053	0.00071	0.0203	0.121	9.652	0.01
Adidas	4928	0.00056	0.0218	0.339	9.397	0.01
Pepsi	8053	0.00045	0.0141	0.361	11.822	0.01
Dick's	5839	0.00111	0.0266	0.321	11.512	0.01
Lululemon	4637	0.00106	0.0309	-0.278	9.326	0.01
S&P 500	8053	0.00040	0.0117	-0.154	10.453	0.01

Note:

Returns computed as simple percentage changes in adjusted closing prices.

Then we need data to measure the sentiment of different players. Thus we use API call to pull NYT article title and abstract textual data. Once we have this text data we then need to give each article a sentiment

rating. We use a trinomial rating system with the following prompt for each text article.

3.2 Article Sentiment

To measure athlete sentiment we require article-level text data. We collect all New York Times articles mentioning Nike-endorsed athletes via the NYT Archive API. For each month from September 1984 through December 2025, we query the API and extract the headline and the abstract for every returned article. The search for these articles is simply each athletes name. The full list and number other articles for each player is below:

Table 2: Article Counts by Athlete (Post-Filtering)

Athlete	Articles
Serena Williams	1133
LeBron James	1105
Roger Federer	1100
Tiger Woods	1059
Rafael Nadal	988
Derek Jeter	780
Kobe Bryant	543
Michael Jordan	466
Maria Sharapova	385
Rory McIlroy	210
Kevin Durant	193
Cristiano Ronaldo	173
Naomi Osaka	137
Neymar	90
Carlos Alcaraz	48
Jannik Sinner	12

For a article to be kept we required the players full name to be in the article to avoid passing mentions. We also removed all articles that mentioned blacklisted word that pertained to war or other extreme news topics. Each article is classified using a trinomial sentiment scale: +1 (good), 0 (neutral), -1 (bad), or NA (irrelevant). The LLM receives the following system prompt, where [Athlete] is replaced with the player’s name:

You are a news article sentiment classifier. You will be given a news article about [Athlete]. Rate how the public would perceive this article for [Athlete]’s reputation and image.

1 = GOOD — positive public perception (win, achievement, positive profile, comeback, milestone, praise)

0 = NEUTRAL — no clear positive or negative public perception (routine update, mixed, informational)

-1 = BAD — negative public perception (loss, scandal, injury, controversy, decline, embarrassment, criticism)

NA = Article is NOT primarily about [Athlete] (only mentions them in passing, or is about a different topic)

Reply with ONLY one of: 1, 0, -1, or NA. No other text.

In the primary specification, each article is rated 25 times using LLaMA 3.2 (3B) via Ollama at a given temperature. We run this procedure at three temperatures: $T = 0.5$, $T = 1.0$, and $T = 2.0$. The article-level sentiment score s_j is the mean across the 25 iterations, and the count of failed or NA responses n_j^{NA} determines article relevance.

As a robustness check we also rate the full texts in a single-pass design using four models at $T = 0.5$: LLaMA 3.2 (3B), LLaMA 3.1 (8B), Mistral 7B, and Gemma 2 (9B). Because each article receives only one rating, the relevance variable is dropped and the regression reduces to four sentiment regressors.

3.3 Merged Example

We see thus given our models and data that we have different tunings of W and Z for each stock and temperature. We show the tuning in the results section as it reveals more robustness to our method but for now we can display a summary of the merged data with a set W and Z that we see later we actually tuned for.

Table 3: Summary Statistics: Athlete Sentiment Ratings (Pre-Merge)

LLM Temp	Variable	N	Mean	Std. Dev.	Skewness	Ex. Kurtosis
T = 0.5						
T = 0.5	Avg. Sentiment	7894	-0.052	0.594	-0.044	-0.679
T = 0.5	N Good	7894	5.126	8.305	1.472	0.590
T = 0.5	N Neutral	7894	13.138	10.001	-0.148	-1.659
T = 0.5	N Bad	7894	6.364	9.585	1.133	-0.466
T = 0.5	N Missing	7894	0.372	1.328	4.342	19.774
T = 1.0						
T = 1.0	Avg. Sentiment	6843	-0.331	0.606	0.423	-1.053
T = 1.0	N Good	6843	4.304	6.592	1.587	1.506
T = 1.0	N Neutral	6843	7.534	7.684	0.695	-0.799
T = 1.0	N Bad	6843	12.014	9.941	0.070	-1.606
T = 1.0	N Missing	6843	1.146	2.345	2.000	2.819
T = 2.0						
T = 2.0	Avg. Sentiment	7197	-0.366	0.688	0.726	-0.804
T = 2.0	N Good	7197	6.711	8.500	0.989	-0.361
T = 2.0	N Neutral	7197	2.056	3.380	1.429	0.901
T = 2.0	N Bad	7197	15.248	8.917	-0.355	-1.205
T = 2.0	N Missing	7197	0.985	2.406	2.262	3.608

Note:

Note: we display sentiment ratings of articles with at least one valid rating

Table 4: Summary Statistics: Post-Merge Dataset by Stock ($T = 0.5$, Stock-Specific Optimal w and z)

Stock	Variable	N	Mean	Std. Dev.	Skewness	Ex. Kurtosis
Nike						
Nike	Return	8052	0.00071	0.02030	0.121	9.662
Nike	Sentiment	8052	-0.06199	0.41905	-0.100	0.057
Nike	Frequency	8052	1.05435	1.34723	2.995	18.103
Nike	Sent. Volatility	8052	0.35469	0.23149	0.096	-0.754
Nike	Relevance	8052	0.93309	0.22359	-3.822	12.990
Nike	No News	8052	0.05266	0.22336	4.005	14.042
Adidas						
Adidas	Return	4927	0.00056	0.02179	0.339	9.395
Adidas	Sentiment	4927	-0.06739	0.51750	-0.102	-0.293
Adidas	Frequency	4927	0.00817	0.01355	3.467	24.423
Adidas	Sent. Volatility	4927	0.13597	0.20732	1.867	3.034
Adidas	Relevance	4927	0.91853	0.24688	-3.305	9.357
Adidas	No News	4927	0.06434	0.24538	3.550	10.606
Pepsi						
Pepsi	Return	8052	0.00045	0.01408	0.361	11.823
Pepsi	Sentiment	8052	-0.03604	0.46964	-0.155	0.291
Pepsi	Frequency	8052	0.04271	0.07383	3.613	27.090
Pepsi	Sent. Volatility	8052	0.13517	0.20599	1.778	2.799
Pepsi	Relevance	8052	0.73482	0.43196	-1.099	-0.767
Pepsi	No News	8052	0.25522	0.43601	1.123	-0.740
Dick's						
Dick's	Return	5838	0.00110	0.02656	0.316	11.545
Dick's	Sentiment	5838	-0.05511	0.48293	-0.160	0.129
Dick's	Frequency	5838	0.00811	0.01314	3.366	23.705
Dick's	Sent. Volatility	5838	0.12696	0.20849	1.967	3.352
Dick's	Relevance	5838	0.79918	0.38802	-1.549	0.448
Dick's	No News	5838	0.18876	0.39135	1.590	0.529
Lululemon						
Lululemon	Return	4636	0.00105	0.03088	-0.279	9.345
Lululemon	Sentiment	4636	-0.05912	0.50344	-0.112	-0.122
Lululemon	Frequency	4636	0.00777	0.01317	3.678	28.073
Lululemon	Sent. Volatility	4636	0.12637	0.20172	1.974	3.519
Lululemon	Relevance	4636	0.87228	0.31343	-2.356	3.713
Lululemon	No News	4636	0.11152	0.31481	2.468	4.090

Note:

Example of merged base data given fixed w and z for each stock (that we tuned for in later sections)

4 Results

4.1 AR(p)-X Tuning

To predict out Nike stock returns we first start with the basics pick how many lags of returns and volatility to include. For now we look at a 50/50 split between AIC and BIC and we see that AR(1) “wins”. This keeps our base simple and makes theoretical sense.

Table 5: Top 10 AR and Volume Lag Specifications (50% AIC + 50% BIC)

AR Lags	Vol Lags	AIC	BIC	AIC Rank	BIC Rank	Combo
1	0	-39883.1	-39862.1	6	2	4.0
0	0	-39882.1	-39868.1	9	1	5.0
2	0	-39883.0	-39855.1	7	4	5.5
3	0	-39883.3	-39848.3	5	7	6.0
4	0	-39885.5	-39843.6	1	11	6.0
5	0	-39883.9	-39835.0	2	16	9.0
4	1	-39883.5	-39834.6	4	17	10.5
1	1	-39881.1	-39853.2	19	5	12.0
4	2	-39883.7	-39827.8	3	22	12.5
1	2	-39881.2	-39846.3	18	8	13.0

Note:

Estimated on Nike daily returns 1994-2026. No sentiment variables included.

4.2 W-Z tuning

Our job now is to add our sentiment variables and see if they have any significant effects on returns. However as mentioned in our model we need to tune for the best W and Z. We do this with 20-fold time-series cross-validation based on expanding windows. We also include rolling windows that drop older training data/years as a robustness check. Once we have these OOS RMSEs we then rank them and take a combined rank of 50% OOS RMSE and 25% AIC and 25% BIC. Obviously AIC and BIC ranks are identical in this case. So we are essentially ranking half on out of sample measures and half on in sample measures.

Table 6: Optimal Decay Parameters by CV Method and LLM Temperature

	Nike		Adidas		Pepsi		Dick's		Lululemon	
	w	z	w	z	w	z	w	z	w	z
Expanding										
T = 0.5	50	0.61	15	0.01	5	0.06	5	0.01	10	0.01
T = 1.0	55	0.57	15	0.01	60	0.84	80	0.05	10	0.51
T = 2.0	60	0.86	15	0.01	5	1.00	25	1.00	70	0.01
Consensus	55	0.63	15	0.01	60	0.88	80	0.01	10	0.01
Rolling (1)										
T = 0.5	50	0.64	5	0.01	5	0.06	65	0.01	5	0.36
T = 1.0	55	0.58	25	0.01	20	0.91	70	1.00	5	0.01
T = 2.0	60	0.83	25	0.01	20	0.90	70	1.00	10	0.01
Consensus	55	0.66	25	0.01	60	0.86	80	0.01	10	0.01
Rolling (2)										
T = 0.5	15	0.31	25	0.01	5	0.01	65	0.01	5	0.36
T = 1.0	15	0.20	25	0.01	5	0.01	80	0.04	10	0.01
T = 2.0	60	0.01	25	0.01	20	1.00	70	0.01	10	0.01
Consensus	15	0.19	25	0.01	5	0.01	80	0.01	10	0.01
Rolling (4)										
T = 0.5	15	0.31	5	0.01	10	0.01	5	0.01	5	0.36
T = 1.0	15	0.20	10	0.01	5	0.01	5	1.00	5	0.01
T = 2.0	65	0.23	10	0.01	20	0.96	5	1.00	10	0.01
Consensus	15	0.20	10	0.01	5	0.01	5	1.00	10	0.01
Rolling (6)										
T = 0.5	95	0.13	5	0.01	100	0.01	65	0.03	100	0.70
T = 1.0	95	0.27	10	0.01	100	1.00	80	0.01	10	0.39
T = 2.0	65	0.33	10	0.01	70	0.86	95	0.57	70	0.01
Consensus	95	0.25	10	0.01	80	0.88	80	0.02	10	0.01
Rolling (8)										
T = 0.5	15	0.29	5	0.01	30	0.01	65	0.01	80	0.27
T = 1.0	15	0.20	10	0.01	20	1.00	80	0.42	10	0.39
T = 2.0	20	0.02	15	0.01	20	0.95	95	0.57	70	0.01
Consensus	15	0.20	15	0.01	20	0.01	80	0.53	100	0.34
Rolling (10)										
T = 0.5	95	0.18	5	0.01	80	0.01	65	0.01	5	0.36
T = 1.0	95	0.05	10	0.01	80	0.01	80	0.01	10	0.01
T = 2.0	65	0.74	10	0.01	70	0.03	95	0.56	10	0.01
Consensus	95	0.08	10	0.01	90	0.01	80	0.01	10	0.01

Note:

w = decay window (days); z = exponential decay rate. Consensus selects minimum mean combo rank across $T \in 0.5, 1.0, 2.0$.

We see consistent W and Z tunes across temperatures which is encouraging. Nike tunes are particularly stable and the values make empirical sense. What is most encouraging is that when we tune the Nike player sentiments on the none-Nike stocks we receive unstable results. Which is our first hint that we are truly

capturing something about Nike and not just some general market phenomenon. Note that the consensus tunes for W and Z are the best across all temperature for that method.

Following this we tune for our “All Players” models that keeps each player as a separate set of sentiment variables. We see similar stability except for the highest temperature window length. However even in that case the decay rate is reasonable.

Table 7: Optimal Decay Parameters by LLM Temperature and CV Method (Nike, Players Model)

	Expanding		Rolling (1)		Rolling (2)		Rolling (4)		Rolling (6)		Rolling (8)		Rolling (10)	
	<i>w</i>	<i>z</i>	<i>w</i>	<i>z</i>	<i>w</i>	<i>z</i>	<i>w</i>	<i>z</i>	<i>w</i>	<i>z</i>	<i>w</i>	<i>z</i>	<i>w</i>	<i>z</i>
T = 0.5	35	0.54	40	0.44	40	0.50	35	0.53	35	0.54	40	0.69	35	0.38
T = 1.0	45	0.50	45	0.50	45	0.50	45	0.50	45	0.50	15	0.52	15	0.40
T = 2.0	5	0.62	45	0.50	5	0.73	5	0.67	5	0.56	45	0.60	45	0.11
Consensus	35	0.50	35	0.50	45	0.50	35	0.50	35	0.50	35	0.57	45	0.15

Note:

w = decay window (days); *z* = exponential decay rate. Consensus row selects minimum mean combo rank across *T* in 0.5, 1.0, 2.0.

Our model that focuses solely on Kobe Bryant’s sentiment variables also shows similar stability across temperatures and methods. However we can note that the windows and decay are generally smaller/more sensitive then the previous results.

Table 8: Optimal Decay Parameters by LLM Temperature and CV Method (Kobe Bryant, Nike)

	Expanding		Rolling (1)		Rolling (2)		Rolling (4)		Rolling (6)		Rolling (8)		Rolling (10)	
	<i>w</i>	<i>z</i>	<i>w</i>	<i>z</i>	<i>w</i>	<i>z</i>	<i>w</i>	<i>z</i>	<i>w</i>	<i>z</i>	<i>w</i>	<i>z</i>	<i>w</i>	<i>z</i>
T = 0.5	70	0.28	70	0.27	70	0.28	70	0.28	70	0.12	70	0.11	60	0.10
T = 1.0	60	0.28	60	0.20	60	0.28	70	0.22	60	0.20	60	0.07	60	0.06
T = 2.0	60	0.23	60	0.19	60	0.23	60	0.07	65	0.05	60	0.06	60	0.06
Consensus	20	0.19	50	1.00	50	1.00	70	0.29	65	0.21	60	0.10	60	0.06

Note:

w = decay window (days); *z* = exponential decay rate. Consensus row selects minimum mean combo rank across *T* in 0.5, 1.0, 2.0.

With all our models tuned for we then can estimate our models. To help interpretationa cross models/temperature we will often favour the consensus W and Z tunings.

4.3 Nike Estimations

First we estimate our main Nike model with the consensus W and Z tuning across temperatures.

Table 9: Nike Full Model: Base, Market Control, and Year FEs (HC3 Standard Errors)

	Base			Market Control			Year FEs		
	T = 0.5	T = 1.0	T = 2.0	T = 0.5	T = 1.0	T = 2.0	T = 0.5	T = 1.0	T = 2.0
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Sentiment	0.0009 (0.0006)	0.0004 (0.0006)	0.0003 (0.0006)	0.0006 (0.0005)	0.0002 (0.0005)	0.0001 (0.0005)	0.0008 (0.0006)	0.0004 (0.0006)	0.0002 (0.0006)
Frequency	-0.0001 (0.0001)	-0.0001 (0.0002)	-0.0001 (0.0002)	-0.0000 (0.0001)	-0.0001 (0.0001)	0.0000 (0.0001)	-0.0002 (0.0002)	-0.0002 (0.0002)	-0.0002 (0.0002)
Sentiment Volatility	0.0024**	0.0025**	0.0019*	0.0022**	0.0025**	0.0018*	0.0029**	0.0027**	0.0022*
Relevance	(0.0011) -0.0073 (0.0059)	(0.0012) -0.0066* (0.0035)	(0.0011) -0.0003 (0.0036)	(0.0010) -0.0031 (0.0045)	(0.0010) -0.0038 (0.0030)	(0.0009) 0.0006 (0.0032)	(0.0013) -0.0105* (0.0063)	(0.0013) -0.0076** (0.0037)	(0.0012) -0.0003 (0.0036)
No News	-0.0044 (0.0058)	-0.0039 (0.0034)	0.0023 (0.0035)	0.0001 (0.0044)	-0.0009 (0.0030)	0.0035 (0.0032)	-0.0091 (0.0063)	-0.0061 (0.0037)	0.0009 (0.0035)
Return (t-1)	-0.0204 (0.0179)	-0.0204 (0.0179)	-0.0202 (0.0179)	-0.0046 (0.0154)	-0.0046 (0.0154)	-0.0044 (0.0154)	-0.0227 (0.0180)	-0.0227 (0.0180)	-0.0223 (0.0180)
Market Return				0.8853*** (0.0236)	0.8852*** (0.0236)	0.8855*** (0.0237)			
AIC	-39907.7	-39907.5	-39904.7	-42353.6	-42353.0	-42351.7	-39864.5	-39863.8	-39859.3
BIC	-39851.7	-39851.5	-39848.8	-42290.7	-42290.0	-42288.8	-39591.7	-39591.1	-39586.5
N	8052	8052	8052	8052	8052	8052	8052	8052	8052
AR(1)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Market Return	No	No	No	Yes	Yes	Yes	No	No	No
Year FEs	No	No	No	No	No	No	Yes	Yes	Yes

Note:

HC3 SEs in parentheses. ***p<0.01, **p<0.05, *p<0.1. Consensus: w = 55, z = 0.63 (expanding CV).

We see that our sentiment volatility variable is positively significant. This is to say that when there is a change/disagreement in the sentiment ratings of Nike players across articles there is a positive return effect. Thus the pure rating of the sentiment is not significant but rather the change in sentiment across articles that is significant. If the sentiment is consistently high or low the markets don't react rather when there is a disagreement they react. We see this finding consistent across yearly FE and market control models that account for any large shocks to the market such as the 2008 financial crisis, COVID-19 pandemic or even Nike specific news enacted in specific years.

Then we are able to estimate our all players model. Note that we only estimated upon players with over 200 articles written about them.

Table 10: Nike Players Model: Sentiment Volatility by Player (HC3 Standard Errors)

	Base			Market Control			Year FEs		
	T = 0.5	T = 1.0	T = 2.0	T = 0.5	T = 1.0	T = 2.0	T = 0.5	T = 1.0	T = 2.0
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Serena Williams									
Sent. Volatility	0.0012 (0.0014)	0.0023* (0.0014)	0.0018 (0.0011)	0.0009 (0.0012)	0.0011 (0.0011)	0.0010 (0.0009)	0.0011 (0.0015)	0.0020 (0.0015)	0.0016 (0.0012)
LeBron James									
Sent. Volatility	0.0007 (0.0013)	0.0017 (0.0012)	0.0011 (0.0012)	0.0003 (0.0010)	0.0008 (0.0010)	0.0002 (0.0010)	0.0004 (0.0014)	0.0016 (0.0013)	0.0012 (0.0012)
Roger Federer									
Sent. Volatility	0.0015 (0.0015)	0.0015 (0.0015)	-0.0000 (0.0013)	0.0007 (0.0012)	0.0013 (0.0012)	0.0004 (0.0010)	0.0012 (0.0015)	0.0010 (0.0015)	-0.0003 (0.0013)
Tiger Woods									
Sent. Volatility	0.0022 (0.0020)	0.0017 (0.0023)	0.0003 (0.0021)	0.0023 (0.0018)	0.0023 (0.0021)	0.0010 (0.0019)	0.0029 (0.0021)	0.0016 (0.0023)	0.0001 (0.0021)
Rafael Nadal									
Sent. Volatility	0.0006 (0.0015)	0.0013 (0.0015)	-0.0007 (0.0012)	-0.0002 (0.0012)	0.0000 (0.0012)	-0.0002 (0.0010)	0.0000 (0.0015)	0.0008 (0.0015)	-0.0010 (0.0012)
Derek Jeter									
Sent. Volatility	0.0010 (0.0016)	0.0027 (0.0018)	0.0034** (0.0016)	0.0014 (0.0013)	0.0029* (0.0015)	0.0035*** (0.0013)	0.0006 (0.0017)	0.0026 (0.0019)	0.0034** (0.0017)
Kobe Bryant									
Sent. Volatility	-0.0061*** (0.0020)	-0.0053*** (0.0020)	-0.0031 (0.0020)	-0.0027* (0.0014)	-0.0026* (0.0015)	-0.0016 (0.0015)	-0.0065*** (0.0021)	-0.0059*** (0.0022)	-0.0034* (0.0020)
Michael Jordan									
Sent. Volatility	0.0052** (0.0025)	0.0068** (0.0031)	0.0063** (0.0031)	0.0031 (0.0022)	0.0050* (0.0027)	0.0041 (0.0027)	0.0062** (0.0027)	0.0067** (0.0033)	0.0064* (0.0033)
Maria Sharapova									
Sent. Volatility	-0.0032 (0.0022)	-0.0024 (0.0021)	-0.0028* (0.0016)	-0.0024 (0.0019)	-0.0014 (0.0017)	-0.0012 (0.0013)	-0.0042* (0.0022)	-0.0034 (0.0022)	-0.0032* (0.0016)
Rory McIlroy									
Sent. Volatility	-0.0034 (0.0030)	-0.0038 (0.0036)	-0.0047* (0.0028)	-0.0022 (0.0022)	-0.0012 (0.0028)	-0.0023 (0.0021)	-0.0026 (0.0031)	-0.0029 (0.0037)	-0.0044 (0.0028)
AIC	-39860.7	-39855.1	-39851.1	-42288.2	-42289.5	-42286.7	-39825.8	-39819.4	-39817.7
BIC	-39490.0	-39484.5	-39480.4	-41910.6	-41911.9	-41909.1	-39238.3	-39231.9	-39230.3
N	8052	8052	8052	8052	8052	8052	8052	8052	8052
AR(1)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sentiment	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Frequency	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Relevance	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No News	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Market Return	No	No	No	Yes	Yes	Yes	No	No	No
Year FEs	No	No	No	No	No	No	Yes	Yes	Yes

Note:

HC3 SEs in parentheses. ***p<0.01, **p<0.05, *p<0.1. w = 35, z = 0.50 (expanding CV).

We see that the sentiment volatility is again significant but not for every player. What is most odd however is that different players have different signs on their coefficient. We first hypothesize that this is due to a difference in reaction to negative ratings. For example, when Kobe Bryant was accused of sexual assault in 2003 individuals would be repulsed from Kobe and thus Nike. Articles during that time would be rated very low. However when Kobe Bryant died articles would again be rated low as his death is undoubtedly bad for Kobe. However, upon his death consumers sympathized with Kobe and thus Nike shown by how quickly his shoe line sold out (Thomas, 2020). We test for this in the following section if this phenomenon could be the cause of changing player signs.

Below we estimate the Kobe model with coefficients for four different periods: pre-Nike contract, during Nike contract (split into two sub-periods), and post-death. We again use the consensus W and Z tuning across temperatures.

Table 11: Kobe Bryant: Career Period Sentiment Effects on Nike Returns (HC3 Standard Errors)

	T = 0.5		T = 1.0		T = 2.0	
	Base	Mkt.	Base	Mkt.	Base	Mkt.
	(1)	(2)	(3)	(4)	(5)	(6)
Sentiment (Pre-Nike)	0.0007 (0.0016)	0.0008 (0.0014)	0.0005 (0.0017)	0.0008 (0.0016)	0.0008 (0.0019)	0.0010 (0.0017)
Sentiment (First Nike Period)	0.0012 (0.0008)	0.0010* (0.0006)	0.0010 (0.0009)	0.0008 (0.0006)	0.0009 (0.0009)	0.0008 (0.0006)
Sentiment (Second Nike Period)	-0.0011 (0.0011)	-0.0011 (0.0011)	-0.0012 (0.0011)	-0.0014 (0.0010)	-0.0012 (0.0012)	-0.0017 (0.0011)
Sentiment (Post-Death)	0.0025 (0.0038)	0.0001 (0.0021)	0.0024 (0.0027)	-0.0004 (0.0016)	0.0013 (0.0030)	-0.0010 (0.0019)
Sent. Volatility (Pre-Nike)	-0.0046 (0.0087)	0.0062 (0.0088)	-0.0142 (0.0124)	-0.0020 (0.0117)	-0.0164 (0.0121)	-0.0071 (0.0109)
Sent. Volatility (First Nike Period)	-0.0060** (0.0026)	-0.0044** (0.0022)	-0.0074** (0.0029)	-0.0049** (0.0024)	-0.0076** (0.0034)	-0.0051* (0.0031)
Sent. Volatility (Second Nike Period)	0.0007 (0.0047)	-0.0029 (0.0047)	-0.0011 (0.0050)	-0.0036 (0.0046)	-0.0009 (0.0047)	-0.0041 (0.0044)
Sent. Volatility (Post-Death)	-0.0785** (0.0341)	-0.0255 (0.0155)	-0.0863*** (0.0331)	-0.0275* (0.0165)	-0.0804* (0.0446)	-0.0217 (0.0206)
AIC	-39900.2	-42330.0	-39902.1	-42330.7	-39897.8	-42330.1
BIC	-39760.3	-42183.1	-39762.2	-42183.8	-39757.9	-42183.3
N	8052	8052	8052	8052	8052	8052
AR(1)	Yes	Yes	Yes	Yes	Yes	Yes
Frequency x Period	Yes	Yes	Yes	Yes	Yes	Yes
Relevance x Period	Yes	Yes	Yes	Yes	Yes	Yes
No News	Yes	Yes	Yes	Yes	Yes	Yes
Market Return	No	Yes	No	Yes	No	Yes

Note:

HC3 SEs in parentheses. ***p<0.01, **p<0.05, *p<0.1. w = 20, z = 0.19.

¹ Pre-Nike: before May 1, 2003. Post-Death: after Jan 26, 2020.

² Nike period split at: Sep 13, 2011.

Obviously we find no significance before Kobe's signing with Nike. Note that for this case we searched for Kobe related articles across the entire return timeframe. Thus it includes articles before and after Kobe's time with Nike. The first period includes Kobe's accusation. We see identical signs in coefficients between the post-death and first period coefficients. This points to the difference in player signs not being due to a difference in reaction to negative news. Thus based on the evidence we are inclined to say that the markets react to disagreement in news differently for different players.

4.4 Single Rating Runs

Up to this point we have run our models using sentiment scores that are the average of 25 runs. Thus we must ask if we get similar results with single shot prompting similar to Lopez-Lira & Tang (2024). While

hardware restraints (mostly ram/vram) limit us from locally running massive LLMs we select three slightly larger LLMs and run then + our old LLM in one shot prompting to see if we get similar results. We show the sentiment ratings summary below. To assist the single run models we keep the temperature = 0.5 to reduce variance in results.

Table 12: Summary Statistics: Single-Run LLM Sentiment Ratings (Nike Athletes)

Model	N	Mean	Std. Dev.	Skewness	Ex. Kurtosis	% Good	% Neutral	% Bad	N Irrelevant
LLaMA 3.2 (3B)	7055	-0.366	0.745	0.706	-0.881	16.1%	31.1%	52.7%	1367
LLaMA 3.1 (8B)	7658	0.144	0.729	-0.228	-1.103	34.8%	44.7%	20.4%	764
Mistral (7B)	8288	0.404	0.642	-0.610	-0.611	49.0%	42.5%	8.6%	134
Gemma 2 (9B)	8169	0.167	0.593	-0.066	-0.338	27.3%	62.1%	10.6%	246

Note:

Sample restricted to articles with a parseable rating. N Irrelevant = articles rated NA (not primarily about the athlete).

We then can tune for the best W and Z same as before.

Table 13: Optimal Decay Parameters and Fit: 25-Run vs Single-Run LLM Models (Nike, Expanding Window CV)

Model	w	z	OOS RMSE	AIC	BIC
25-Run (T = 0.5)	55	0.63	0.019798	-39907.7	-39851.7
LLaMA 3.2 (3B)	15	0.16	0.019622	-39905.0	-39856.0
LLaMA 3.1 (8B)	85	0.31	0.019628	-39910.6	-39861.7
Mistral (7B)	95	1.00	0.019608	-39904.7	-39855.7
Gemma 2 (9B)	55	0.13	0.019638	-39907.2	-39858.3

Note:

Combo rank: 50% RMSE, 25% AIC, 25% BIC. All models T = 0.5.

We dont see any real differences in terms of RMSE between the models. The differences in the single vs 25-run models is surely due to the NA variables dropping articles/ the extra relevance variable that is not designed for forecasting but rather for the interpretability of the 25 run models.

Finally we can estimate these models.

Table 14: Single-Run LLM Models: Base and Market Control Specifications (HC3 Standard Errors)

	LLaMA 3.2 (3B)		LLaMA 3.1 (8B)		Mistral (7B)		Gemma 2 (9B)	
	Base	Mkt.	Base	Mkt.	Base	Mkt.	Base	Mkt.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Sentiment	-0.0001 (0.0004)	-0.0002 (0.0003)	0.0008* (0.0004)	0.0003 (0.0003)	-0.0013 (0.0012)	-0.0013 (0.0011)	0.0006 (0.0005)	0.0005 (0.0004)
Frequency	-0.0010 (0.0012)	0.0003 (0.0010)	-0.0002 (0.0005)	0.0002 (0.0004)	0.0000 (0.0000)	0.0000** (0.0000)	-0.0011 (0.0013)	0.0004 (0.0011)
Sentiment Volatility	0.0021** (0.0009)	0.0012 (0.0008)	0.0017* (0.0009)	0.0017** (0.0008)	-0.0012 (0.0014)	-0.0008 (0.0013)	0.0017 (0.0011)	0.0015 (0.0009)
No News	0.0014* (0.0008)	0.0013* (0.0007)	0.0026*** (0.0008)	0.0026*** (0.0008)	0.0018 (0.0014)	0.0023* (0.0013)	0.0024*** (0.0008)	0.0025*** (0.0008)
Return (t-1)	-0.0200 (0.0178)	-0.0040 (0.0154)	-0.0201 (0.0179)	-0.0043 (0.0154)	-0.0200 (0.0179)	-0.0043 (0.0154)	-0.0198 (0.0179)	-0.0041 (0.0154)
Market Return		0.8849*** (0.0237)		0.8851*** (0.0237)		0.8856*** (0.0237)		0.8854*** (0.0237)
AIC	-39905.0	-42346.4	-39910.6	-42354.8	-39904.7	-42352.1	-39907.2	-42353.2
BIC	-39856.0	-42290.4	-39861.7	-42298.8	-39855.7	-42296.1	-39858.3	-42297.2
N	8052	8052	8052	8052	8052	8052	8052	8052
AR(1)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Market Return w	No 15	Yes 15	No 85	Yes 85	No 95	Yes 95	No 55	Yes 55
z	0.16	0.16	0.31	0.31	1.00	1.00	0.13	0.13

Note:

***p<0.01, **p<0.05, *p<0.1. w and z are model-specific optimal decay parameters from expanding window CV.

And we see that we have poor significance across the single run models. This is likely due to the noise of the W and Z selection process. If we force all model to have the same w and z as the 25-run model we get similar results (see appendix). This implies the need for multiple runs to get stable sentiment volatility estimates.

5 Final Remarks

5.1 Summary

Revisiting our research questions, we see that we have successfully supplied the first daily link between player performance/the public view and the stock returns of that players sponsor. We find that sentiment volatility is the most robust predictor of returns across players and model specifications. This implies that disagreement in news about a player is more important than the average sentiment of news about a player. We also find that the market reacts differently to disagreement for different players. We see cases of Kobe and Sharapova having negative sentiment volatility coefficients while players like Michael Jordan and Jeter have positive coefficients. The difference between these two player types clearing being controversy as Kobe had a very public sexual assault and Sharapova had a very public doping scandal. We also show proof that

single LLM runs do not provide enough heterogeneity in this case to get significant and robust results.

5.2 Limitations and Future Research

During our research we run into multiple limitations that simply need more time and computational power to check. First and most easily we should run the 25-run models on the 3 chosen models. We could also try rotating between multiple models among the runs. The number of 25 could also be played with.

Multiple news sources should be employed. Especially European ones as we lack results/articles for soccer players. On a similar note a expansion of the players included would be useful. While we selected some of the most popular athletes with Nike it would be interesting to examine if less famous players have similar effects. An important note is that with the current set up we could not split the main model by gender due to lack of heterogeneity among top-tier Nike endorsers where only three of sixteen athletes in our sample are female, all from tennis. This will certainly change as more female athletes gain major endorsement deals. In the interim, a gender analysis could still be conducted among mid-tier contracted athletes where the gender balance is more even.”

6 Acknowledgments

LLM’s such as Opus 4.5, Sonnet 4.6 and Opus 4.6 where used to assist the coding as well as catching grammar mistakes for this project.

7 References

- Agrawal, J. & Kamakura, W. A. (1995). The Economic Worth of Celebrity Endorsers: An Event Study Analysis. *Journal of Marketing*, 59(3), 56–62. <https://doi.org/10.1177/002224299505900305>
- Chung, K. Y., Derdenger, T. P. & Srinivasan, K. (2013). Economic Value of Celebrity Endorsements: Tiger Woods’ Impact on Sales of Nike Golf Balls. *Marketing Science*, 32(2), 271–293. <https://doi.org/10.1287/mksc.1120.0760>
- Elberse, A. & Verleun, J. (2012). The Economic Value of Celebrity Endorsements. *Journal of Advertising Research*, 52(2), 149–165. <https://doi.org/10.2501/JAR-52-2-149-165>
- Hood, M. (2012). The Tiger Woods Scandal: A Cautionary Tale for Event Studies. *Managerial Finance*, 38(5), 543–558. <https://doi.org/10.1108/03074351211217850>
- Kirtac, K. & Germano, G. (2024). Sentiment Trading with Large Language Models. *Finance Research Letters*, 62, 105227. <https://doi.org/10.1016/j.frl.2024.105227>
- Knittel, C. R. & Stango, V. (2014). Celebrity Endorsements, Firm Value, and Reputation Risk: Evidence from the Tiger Woods Scandal. *Management Science*, 60(1), 21–37. <https://doi.org/10.1287/mnsc.2013.1749>
- Kunkel, T. (2023). How Michael Jordan Revolutionized the Sneaker Industry. Temple University, School of Sport, Tourism and Hospitality Management. Retrieved from <https://now.temple.edu/news/2023-04-03/how-michael-jordan-revolutionized-sneaker-industry-our-relationship-shoes>
- Lopez-Lira, A. & Tang, Y. (2024). Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models. Working Paper, University of Florida. <https://arxiv.org/abs/2304.07619>
- Tetlock, P. C. (2007). Giving Content to Investor Sentiment: The Role of Media in the Stock Market. *Journal of Finance*, 62(3), 1139–1168. <https://doi.org/10.1111/j.1540-6261.2007.01232.x>
- Thomas, L. (2020, January 28). Nike Sells Out of Kobe Bryant Merchandise Online After Death. *CNBC*. Retrieved from <https://www.cnbc.com/2020/01/28/nike-sells-out-of-kobe-bryant-merchandise-online-after-death.html>

8 Appendix

See estimations of our models that run Nike athlete sentiment on other brands below. Note the lack of significance below which points to use capturing the correct effect rather than a sector wide effect.

Table 15: Full Model: HC3 t-statistics, Non-Nike Stocks (Stock-Specific Consensus w and z)

	Base			Market Control			Year FEs		
	T = 0.5	T = 1.0	T = 2.0	T = 0.5	T = 1.0	T = 2.0	T = 0.5	T = 1.0	T = 2.0
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Adidas									
Sentiment	0.892	1.629	1.893*	-0.108	0.814	0.857	0.653	1.346	1.654*
Frequency	-1.830*	-1.876*	-1.812*	-1.453	-1.414	-1.201	-2.053**	-2.139**	-2.030**
Sentiment Volatility	-0.356	-0.839	-0.789	0.040	-1.134	-1.180	-0.491	-0.899	-0.851
Relevance	-0.916	0.218	-0.475	-0.136	0.687	-0.824	-1.071	0.069	-0.572
No News	-0.964	0.073	-0.693	-0.208	0.406	-1.102	-1.037	0.130	-0.613
Return (t-1)	0.206	0.197	0.201	1.343	1.328	1.340	0.019	0.014	0.018
Market Return				26.293***	26.298***	26.267***			
AIC	-23713.5	-23715.8	-23715.9	-25675.8	-25679.4	-25679.1	-23696.7	-23698.8	-23698.5
BIC	-23661.5	-23663.8	-23663.8	-25617.3	-25620.9	-25620.6	-23521.1	-23523.2	-23522.9
N	4927	4927	4927	4927	4927	4927	4927	4927	4927
w	15	15	15	15	15	15	15	15	15
z	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
AR(1)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Market Return	No	No	No	Yes	Yes	Yes	No	No	No
Year FEs	No	No	No	No	No	No	Yes	Yes	Yes
Pepsi									
Sentiment	-0.959	-0.395	0.325	-1.074	-0.753	0.135	-1.217	-0.472	0.453
Frequency	0.493	1.051	0.987	0.818	1.146	1.301	0.534	1.044	0.811
Sentiment Volatility	-0.759	-1.420	-1.607	-0.511	-0.637	-1.180	-0.642	-1.129	-1.565
Relevance	1.183	0.783	2.193**	1.930*	1.204	2.508**	1.160	0.676	2.132**
No News	1.061	0.384	1.852*	1.866*	0.982	2.378**	0.976	0.253	1.595
Return (t-1)	-3.641***	-3.648***	-3.652***	-3.639***	-3.641***	-3.652***	-3.709***	-3.715***	-3.719***
Market Return				28.434***	28.435***	28.437***			
AIC	-45855.5	-45856.3	-45857.5	-47828.8	-47826.9	-47829.8	-45811.4	-45811.7	-45812.8
BIC	-45799.5	-45800.4	-45801.5	-47765.8	-47764.0	-47766.8	-45538.7	-45538.9	-45540.0
N	8052	8052	8052	8052	8052	8052	8052	8052	8052
w	60	60	60	60	60	60	60	60	60
z	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88
AR(1)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Market Return	No	No	No	Yes	Yes	Yes	No	No	No
Year FEs	No	No	No	No	No	No	Yes	Yes	Yes
Dick's									
Sentiment	1.494	1.755*	1.636	0.766	0.998	0.721	1.869*	2.258**	2.024**
Frequency	-0.305	-1.262	-0.591	0.617	-0.450	0.492	-0.375	-1.297	-0.657
Sentiment Volatility	0.066	1.245	-0.057	0.161	1.253	-0.341	0.329	1.713*	0.547
Relevance	-2.293**	-0.861	-0.609	-1.962**	-0.778	-0.786	-2.181**	-0.844	-0.530
No News	-0.752	0.635	0.458	-0.479	0.504	0.230	-1.011	0.358	0.079
Return (t-1)	0.403	0.418	0.446	0.697	0.713	0.739	0.218	0.211	0.246
Market Return				25.599***	25.631***	25.626***			
AIC	-25793.4	-25791.2	-25786.6	-27460.7	-27458.6	-27455.1	-25766.8	-25766.8	-25761.2
BIC	-25740.0	-25737.9	-25733.2	-27400.6	-27398.5	-27395.0	-25559.9	-25560.0	-25554.3
N	5838	5838	5838	5838	5838	5838	5838	5838	5838
w	80	80	80	80	80	80	80	80	80
z	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
AR(1)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Market Return	No	No	No	Yes	Yes	Yes	No	No	No
Year FEs	No	No	No	No	No	No	Yes	Yes	Yes
Lululemon									
Sentiment	1.837*	2.169**	2.025**	1.148	1.537	1.220	1.897*	2.356**	2.219**
Frequency	-1.182	-1.537	-1.277	-0.933	-1.290	-0.868	-1.486	-1.816*	-1.572
Sentiment Volatility	-0.193	0.118	-0.353	0.113	0.089	-0.502	-0.115	0.440	-0.043
Relevance	-1.927*	-0.694	0.006	-1.199	-0.368	-0.166	-1.941*	-0.655	-0.083
No News	-2.232**	-1.336	-0.614	-1.562	-1.133	-0.852	-2.130**	-1.174	-0.561
Return (t-1)	1.109	1.132	1.134	2.344**	2.350**	2.347**	0.802	0.823	0.824
Market Return				21.753***	21.785***	21.786***			
AIC	-19087.1	-19086.7	-19085.4	-20490.9	-20493.2	-20491.4	-19078.9	-19079.1	-19077.4
BIC	-19035.5	-19035.1	-19033.8	-20432.9	-20435.2	-20433.4	-18911.4	-18911.7	-18909.9
N	4636	4636	4636	4636	4636	4636	4636	4636	4636
w	10	10	10	10	10	10	10	10	10
z	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
AR(1)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Market Return	No	No	No	Yes	Yes	Yes	No	No	No
Year FEs	No	No	No	No	No	No	Yes	Yes	Yes

Note:

HC3 t-statistics. ***p<0.01, **p<0.05, *p<0.1. Stock-specific consensus w and z from expanding window CV.

Table 16: Full Model: HC3 t-statistics, All Non-Nike Stocks (Nike Consensus w and z)

	Base			Market Control			Year FEs		
	T = 0.5	T = 1.0	T = 2.0	T = 0.5	T = 1.0	T = 2.0	T = 0.5	T = 1.0	T = 2.0
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Adidas									
Sentiment	0.914	2.137**	1.802*	0.044	1.664*	1.079	0.657	1.891*	1.699*
Frequency	-0.791	-0.635	-0.772	-0.329	-0.323	-0.359	-0.838	-0.751	-0.861
Sentiment Volatility	0.662	-0.326	0.132	0.683	-0.255	0.486	0.442	-0.425	-0.108
Relevance	0.127	-0.294	-0.806	1.003	0.790	-0.672	0.064	-0.182	-0.731
No News	-0.134	-0.602	-0.815	0.598	0.319	-0.477	-0.243	-0.545	-0.685
Return (t-1)	0.229	0.213	0.211	1.357	1.334	1.335	0.055	0.040	0.038
Market Return				26.293***	26.290***	26.239***			
AIC	-23709.4	-23713.3	-23713.0	-25675.7	-25677.9	-25676.3	-23691.0	-23694.9	-23694.5
BIC	-23657.4	-23661.3	-23660.9	-25617.2	-25619.4	-25617.8	-23515.4	-23519.4	-23518.9
N	4927	4927	4927	4927	4927	4927	4927	4927	4927
AR(1)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Market Return	No	No	No	Yes	Yes	Yes	No	No	No
Year FEs	No	No	No	No	No	No	Yes	Yes	Yes
Pepsi									
Sentiment	-0.514	-0.244	0.078	-0.976	-0.688	-0.196	-0.677	-0.353	0.123
Frequency	-0.491	0.458	0.009	0.101	0.992	0.711	-0.359	0.486	0.041
Sentiment Volatility	0.596	-0.657	-0.192	0.506	-0.687	-0.296	0.786	-0.558	-0.151
Relevance	0.986	0.514	1.662*	1.790*	1.319	2.052**	0.949	0.476	1.599
No News	0.892	0.230	1.533	1.742*	1.103	2.025**	0.703	0.085	1.226
Return (t-1)	-3.637***	-3.635***	-3.642***	-3.636***	-3.634***	-3.641***	-3.708***	-3.702***	-3.708***
Market Return				28.430***	28.441***	28.430***			
AIC	-45854.6	-45853.4	-45854.4	-47828.6	-47826.5	-47827.2	-45810.5	-45808.7	-45809.2
BIC	-45798.7	-45797.5	-45798.5	-47765.6	-47763.5	-47764.2	-45537.7	-45535.9	-45536.5
N	8052	8052	8052	8052	8052	8052	8052	8052	8052
AR(1)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Market Return	No	No	No	Yes	Yes	Yes	No	No	No
Year FEs	No	No	No	No	No	No	Yes	Yes	Yes
Dick's									
Sentiment	0.826	1.203	1.441	0.222	0.726	1.023	1.367	1.846*	1.904*
Frequency	-0.116	-0.665	-0.302	0.540	-0.223	0.340	-0.136	-0.606	-0.315
Sentiment Volatility	-0.075	0.464	-0.816	-0.239	0.477	-0.869	0.146	0.902	-0.206
Relevance	-2.135**	-1.874*	-0.711	-1.809*	-1.334	-0.538	-2.011**	-1.790*	-0.660
No News	-1.773*	-1.445	-1.175	-1.423	-0.968	-1.007	-1.991**	-1.712*	-1.421
Return (t-1)	0.438	0.457	0.479	0.729	0.751	0.761	0.243	0.231	0.261
Market Return				25.645***	25.690***	25.625***			
AIC	-25789.7	-25788.4	-25786.3	-27458.4	-27455.7	-27455.6	-25764.0	-25765.3	-25762.3
BIC	-25736.4	-25735.1	-25733.0	-27398.3	-27395.6	-27395.6	-25557.2	-25558.5	-25555.5
N	5838	5838	5838	5838	5838	5838	5838	5838	5838
AR(1)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Market Return	No	No	No	Yes	Yes	Yes	No	No	No
Year FEs	No	No	No	No	No	No	Yes	Yes	Yes
Lululemon									
Sentiment	1.644	1.548	1.580	1.014	1.007	0.932	1.639	1.761*	1.809*
Frequency	-0.335	-0.222	-0.172	0.035	0.003	0.153	-0.777	-0.640	-0.669
Sentiment Volatility	1.506	0.760	-0.048	1.651*	1.050	0.304	1.107	0.489	-0.262
Relevance	-1.280	-1.081	0.281	-0.628	-0.079	0.608	-1.214	-1.109	0.191
No News	-0.028	-0.787	0.612	0.633	0.074	1.123	-0.080	-0.788	0.588
Return (t-1)	1.100	1.142	1.146	2.329**	2.370**	2.365**	0.798	0.822	0.820
Market Return				21.690***	21.684***	21.671***			
AIC	-19084.4	-19080.8	-19080.5	-20491.5	-20487.0	-20487.6	-19075.6	-19073.7	-19074.2
BIC	-19032.9	-19029.3	-19029.0	-20433.5	-20429.0	-20429.7	-18908.1	-18906.2	-18906.7
N	4636	4636	4636	4636	4636	4636	4636	4636	4636
AR(1)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Market Return	No	No	No	Yes	Yes	Yes	No	No	No
Year FEs	No	No	No	No	No	No	Yes	Yes	Yes

Note:

HC3 t-statistics. ***p<0.01, **p<0.05, *p<0.1. Nike consensus w = 55, z = 0.63 applied to all stocks (expanding CV).

Table 17: Single-Run LLM Models: Forced 25-Run Consensus Decay Parameters (HC3 Standard Errors)

	LLaMA 3.2 (3B)		LLaMA 3.1 (8B)		Mistral (7B)		Gemma 2 (9B)	
	Base	Mkt.	Base	Mkt.	Base	Mkt.	Base	Mkt.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Sentiment	-0.0001 (0.0005)	-0.0002 (0.0004)	0.0010* (0.0005)	0.0004 (0.0004)	0.0000 (0.0006)	-0.0003 (0.0005)	0.0008 (0.0006)	0.0007 (0.0005)
Frequency	-0.0001 (0.0002)	0.0000 (0.0001)	-0.0000 (0.0002)	0.0000 (0.0001)	-0.0000 (0.0001)	0.0000 (0.0001)	-0.0001 (0.0001)	-0.0000 (0.0001)
Sentiment Volatility	0.0012 (0.0009)	0.0012 (0.0008)	0.0015 (0.0010)	0.0017* (0.0009)	0.0009 (0.0011)	0.0012 (0.0010)	0.0017 (0.0011)	0.0017* (0.0010)
No News	0.0024** (0.0011)	0.0028*** (0.0010)	0.0026*** (0.0009)	0.0026*** (0.0008)	0.0021** (0.0011)	0.0024** (0.0010)	0.0027*** (0.0009)	0.0028*** (0.0008)
Return (t-1)	-0.0198 (0.0179)	-0.0041 (0.0154)	-0.0199 (0.0179)	-0.0041 (0.0154)	-0.0196 (0.0179)	-0.0040 (0.0154)	-0.0198 (0.0179)	-0.0041 (0.0154)
Market Return		0.8856*** (0.0237)		0.8850*** (0.0237)		0.8858*** (0.0237)		0.8853*** (0.0237)
AIC	-39902.6	-42349.2	-39908.7	-42351.9	-39901.5	-42348.6	-39907.4	-42354.0
BIC	-39853.7	-42293.2	-39859.8	-42295.9	-39852.5	-42292.7	-39858.4	-42298.0
N	8052	8052	8052	8052	8052	8052	8052	8052
AR(1)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Market Return	No	Yes	No	Yes	No	Yes	No	Yes

Note:

HC3 SEs in parentheses. ***p<0.01, **p<0.05, *p<0.1. All models use 25-run consensus w = 55, z = 0.63 (expanding CV).

We re-run the main model on only data after the LLaMA 3.2 (3B) training cutoff of December 2023 plus a buffer of trading days so no pre-cutoff articles leak into the decay window. We still find significance on sentiment volatility at the Temperature = 0.5 despite larger standard errors from the smaller sample which suggests our results are not driven by the model having possibly seen the articles during training.

Table 18: Information Leakage Check: Nike Model on Post-Cutoff Data Only (HC3 Standard Errors)

	T = 0.5		T = 1.0		T = 2.0	
	Base	Mkt.	Base	Mkt.	Base	Mkt.
	(1)	(2)	(3)	(4)	(5)	(6)
Sentiment	0.0016 (0.0031)	0.0015 (0.0028)	-0.0006 (0.0027)	0.0006 (0.0023)	0.0012 (0.0034)	0.0024 (0.0031)
Frequency	0.0011 (0.0029)	0.0017 (0.0026)	0.0028 (0.0041)	0.0037 (0.0038)	0.0029 (0.0042)	0.0031 (0.0038)
Sentiment Volatility	0.0094*	0.0093**	0.0016	0.0005	0.0024	0.0004
	(0.0048)	(0.0045)	(0.0039)	(0.0034)	(0.0052)	(0.0047)
Relevance	-0.0102 (0.0095)	-0.0055 (0.0097)	-0.0100 (0.0125)	-0.0056 (0.0110)	0.0083 (0.0127)	0.0131 (0.0120)
No News	-0.0090 (0.0105)	-0.0037 (0.0105)	-0.0092 (0.0109)	-0.0058 (0.0101)	0.0038 (0.0129)	0.0083 (0.0126)
Return (t-1)	-0.0839 (0.0514)	-0.0718 (0.0638)	-0.0786 (0.0522)	-0.0659 (0.0641)	-0.0816 (0.0519)	-0.0697 (0.0640)
Market Return		0.8984*** (0.1909)		0.9002*** (0.1891)		0.9093*** (0.1880)
AIC	-2139.0	-2211.4	-2135.7	-2207.5	-2136.6	-2210.2
BIC	-2105.9	-2174.1	-2102.6	-2170.3	-2103.4	-2172.9
N	465	465	465	465	465	465
AR(1)	Yes	Yes	Yes	Yes	Yes	Yes
Market Return	No	Yes	No	Yes	No	Yes

Note:

HC3 SEs in parentheses. ***p<0.01, **p<0.05, *p<0.1. Consensus w = 55, z = 0.63. LLaMA 3.2 (3B) training data cutoff: December 2023. Sample restricted to trading days from Feb 26, 2024 to Dec 31, 2025 (N = 465).

The quantizations of the LLMs are shown below:

Table 19: LLM Model Specifications (Ollama)

Model	Parameters	Quantization	Context Length
LLaMA 3.2	3.2B	Q4_K_M	131,072
LLaMA 3.1	8.0B	Q4_K_M	131,072
Mistral	7.2B	Q4_K_M	32,768
Gemma 2	9.2B	Q4_0	8,192